

Cell-type-specific origins of spinal rhythmicity at different locomotor speeds in larval zebrafish

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

[About eLife's process](#)

Reviewed preprint version 1

February 7, 2024 (this version)

Posted to preprint server

January 12, 2024

Sent for peer review

December 1, 2023

Moneeza A. Agha, Sandeep Kishore, David L. McLean 

Department of Neurobiology, Northwestern University, Evanston, IL 60201 • Interdisciplinary Biological Sciences Graduate Program, Northwestern University, Evanston, IL 60201

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

Different speeds of locomotion require heterogeneous spinal populations, but a common mode of rhythm generation is presumed to exist. Here, we explore the cellular versus synaptic origins of spinal rhythmicity at different speeds by performing electrophysiological recordings from premotor excitatory interneurons in larval zebrafish. Chx10-labeled V2a neurons are divided into at least two subtypes proposed to play distinct roles in timing and intensity control. Consistent with distinct rhythm generating and output patterning functions within the spinal V2a population, we find that one subtype is recruited exclusively at slow or fast speeds and exhibits intrinsic cellular properties suitable for rhythmogenesis at those speeds, while the other subtype is recruited more reliably at all speeds and lacks appropriate rhythmogenic cellular properties. Unexpectedly, however, phasic firing patterns during locomotion in rhythmogenic and non-rhythmogenic subtypes are best explained by distinct modes of synaptic inhibition linked to cell-type and speed. At fast speeds reciprocal inhibition in rhythmogenic V2a neurons supports phasic firing, while recurrent inhibition in non-rhythmogenic V2a neurons helps pattern motor output. In contrast, at slow speeds recurrent inhibition in rhythmogenic V2a neurons supports phasic firing, while non-rhythmogenic V2a neurons rely on reciprocal inhibition alone to pattern output. Our findings suggest cell-type-specific, not common, modes of rhythmogenesis generate and coordinate different speeds of locomotion.

eLife assessment

The manuscript by Agha et al. provides a **fundamental** understanding regarding the participation of V2a interneurons in generating and patterning the locomotor rhythm. The authors provide **convincing** and **solid** evidence regarding the heterogeneity of V2a neurons in their intrinsic and synaptic properties and how these shape their outputs. The manuscript could be much improved by the inclusion of statistical analysis of some of the key data currently presented qualitatively.

Introduction

All forms of vertebrate locomotion rely on rhythmic activity in muscles operating as functional antagonists (Grillner, 1985 [↗](#)). For instance, swimming in fishes relies on left-right alternation of axial muscles across the body, while walking in mammals relies on flexor-extensor alternation along it. A landmark hypothesis posed over a century ago argued for corresponding excitatory ‘half-centers’ within spinal cord (e.g., left-right or flexor-extensor), with reciprocal, feedforward inhibition between half-centers generating rhythmic output (Brown, 1911 [↗](#)). Since then reciprocal inhibition has been relegated to patterning activity between so-called ‘unit burst generators’ capable of rhythmogenesis via recurrent, feedback excitatory connections (Grillner and Kozlov, 2021 [↗](#)).

The idea of solitary unit burst generators reliant on recurrent connectivity disconnects rhythmogenesis from activity in other unit burst generators, which helps explain the flexible patterns of coordination necessary to effectively navigate through unpredictable environments. For example, simulations adjusting the synaptic coupling strength between segmental axial unit burst generators can explain changes in the speed and direction of swimming in lampreys (Kozlov et al., 2009 [↗](#)). Likewise, simulations altering the synaptic coupling between segmental limb unit burst generators can replicate different speeds and gaits of locomotion in cats (Grillner, 1981 [↗](#)).

Another theory explaining speed-dependent changes in coordination stratifies ‘rhythm’ and ‘pattern’ control (McCrea and Rybak, 2008 [↗](#)), where populations of interneurons generating higher or lower frequency rhythms feed into separate populations coordinating distinct patterns of motor output (Ausborn et al., 2019 [↗](#)). In support, studies in zebrafish and mice suggest separate spinal populations are active at different locomotor speeds (McLean et al., 2008 [↗](#); Talpalar et al., 2013 [↗](#); Satou et al., 2020 [↗](#); Picton et al., 2022 [↗](#)). Critically, however, a common mode of rhythmogenesis is presumed to exist at all speeds regardless of topological framework, reliant on either cellular properties or synaptic drive (Guertin, 2009 [↗](#); Brocard et al., 2010 [↗](#); Ryczko et al., 2010 [↗](#); Ziskind-Conhaim and Hochman, 2017 [↗](#); El Manira, 2023 [↗](#)).

Here, we have explored the origins of spinal rhythmogenesis at different speeds using the larval zebrafish model system, where it is possible to link the identity, physiology and connectivity of interneurons to their recruitment patterns at different speeds of locomotion. Like all fish, larval zebrafish generate two basic modes of forward swimming or ‘gaits’ (Di Santo et al., 2021 [↗](#)). Anguilliform mode is typically reserved for escaping predators and uses higher frequency/amplitude oscillations of the trunk, while carangiform mode utilizes lower frequency/amplitude oscillations of the tail for exploration and pursuing prey (Budick and O’Malley, 2000 [↗](#); Muller and van Leeuwen, 2004 [↗](#); Thorsen et al., 2004 [↗](#); Patterson et al., 2013 [↗](#); Bhattacharyya et al., 2021 [↗](#)).

Swimming at different speeds is generated by axial motor neurons organized into fast, slow and intermediate units distinguished by birth order (Myers et al., 1986 [↗](#)), size (McLean et al., 2007 [↗](#)), relative innervation of fast and slow muscle fibers (Bello-Rojas et al., 2019 [↗](#)), and molecular signatures (D’Elia et al., 2023 [↗](#)). Current-clamp and voltage-clamp recordings from motor neurons have revealed differences in rhythmogenic cellular properties and synaptic drive between the different types (Menelaou and McLean, 2012 [↗](#); Kishore et al., 2014 [↗](#)), suggesting different forms of rhythmogenesis could be responsible for different speeds of locomotion. To test this idea, we performed current-clamp and voltage-clamp recordings from Chx10-labeled spinal V2a interneurons, which provide a major source of phasic premotor excitation during locomotion in zebrafish and mice (McLean and Dougherty, 2015 [↗](#)).

We find that V2a neurons sort into rhythm-generating and pattern-forming subtypes based on distinct cellular properties linked to distinct patterns of recruitment and synaptic drive. Critically, among V2a neurons that qualify as rhythm-generating, reciprocal inhibition originating from

contralateral sources supports phasic firing patterns at fast speeds, while recurrent inhibition from ipsilateral sources supports phasic firing patterns at slow speeds. For V2a neurons considered pattern-forming, the opposite is true – recurrent inhibition supports phasic firing at fast speeds and reciprocal inhibition at slow speeds. Ultimately, our findings reveal unexpected cell-type-specific and speed-dependent complexity in the mechanisms of spinal rhythmogenesis underlying locomotion.

Results

Identifying V2a neurons for recordings during ‘fictive’ escape swimming

From our previous work studying V2a neurons in larval zebrafish we know they are organized into: 1) fast and slow subtypes based on dorso-ventral soma position and birth order (McLean et al., 2007 [↗](#); McLean et al., 2008 [↗](#); McLean and Fetcho, 2009 [↗](#)), and 2) distinct subtypes that provide stronger direct versus indirect control of motor neuron recruitment (Menelaou et al., 2014 [↗](#); Menelaou and McLean, 2019 [↗](#)). Specifically, V2a neurons with descending axons (V2a-D) provide stronger inputs to other premotor interneurons, while V2a neurons with bifurcating axons (V2a-B) provide stronger inputs to motor neurons (Menelaou and McLean, 2019 [↗](#)), in line with differential rhythm and pattern control (Feldman, 1986 [↗](#)). However, we do not know if the same mechanisms of rhythmogenesis shape the phasic activation of distinct V2a types at different speeds.

To record from distinct types of V2a neurons, we targeted GFP-positive neurons located at midbody in the Tg(Chx10:GFP) line (Fig. 1A,B [↗](#)) (Kimura et al., 2006 [↗](#)), since midbody neurons would be recruited during both fast anguilliform and slow carangiform modes (Muller and van Leeuwen, 2004 [↗](#); Thorsen et al., 2004 [↗](#)). V2a-D and V2a-B neurons recruited at fast speeds can be distinguished by levels of Chx10 expression in transgenic lines (Menelaou and McLean, 2019 [↗](#)), as reported in mice (Hayashi et al., 2018 [↗](#)). To get a better sense of differences in the number and spatial distribution of distinct types, we split GFP-positive neurons into two categories based on a bimodal distribution of fluorescence intensity and normalized their soma positions to dorso-ventral and medio-lateral axes in the spinal cord. Larger, dimmer V2a neurons intermingled with smaller, brighter neurons along the dorso-ventral axis (Fig. 1C [↗](#), left), but occupied more lateral positions (Fig. 1C [↗](#), right). Statistical analysis confirmed these observations, with no significant difference in either dorso-ventral distribution (Fig. 1D [↗](#), left) or number per segment between the low or high GFP-positive types (Fig. 1D [↗](#), right), but a significant difference in mediolateral position (Fig. 1E [↗](#)).

To evaluate firing patterns and synaptic drive responsible for rhythmogenesis in different types, we recorded from V2a neurons during ‘fictive’ swimming evoked by a mild electrical shock, while simultaneously monitoring rhythmic motor output using ventral rootlet recordings (Fig. 1F [↗](#)). During real swimming, escape responses decelerate, beginning in fast anguilliform mode and ending in slow carangiform mode (McLean et al., 2008 [↗](#); McLean and Fetcho, 2009 [↗](#)). Plots of ventral rootlet burst frequencies with time confirm this pattern during ‘fictive’ escape responses, where we observe the highest frequencies at the beginning of the swim episode immediately following the stimulus (Fig. 1G [↗](#), left). Notably, zebrafish swimming is episodic (McHenry and Lauder, 2006 [↗](#); Soto and McHenry, 2020 [↗](#)) and so it can continue intermittently after the initial escape swim episode. Subsequent episodes (swim 2+) rarely exceeded peak firing frequencies observed in the first stimulus-evoked episode (> 60 Hz in swim 1) and there was more variability in frequency (Fig. 1G [↗](#), right). For purposes of analysis we pooled data from the first (swim 1) and subsequent (swim 2+) episodes to capture as broad a frequency range as possible.

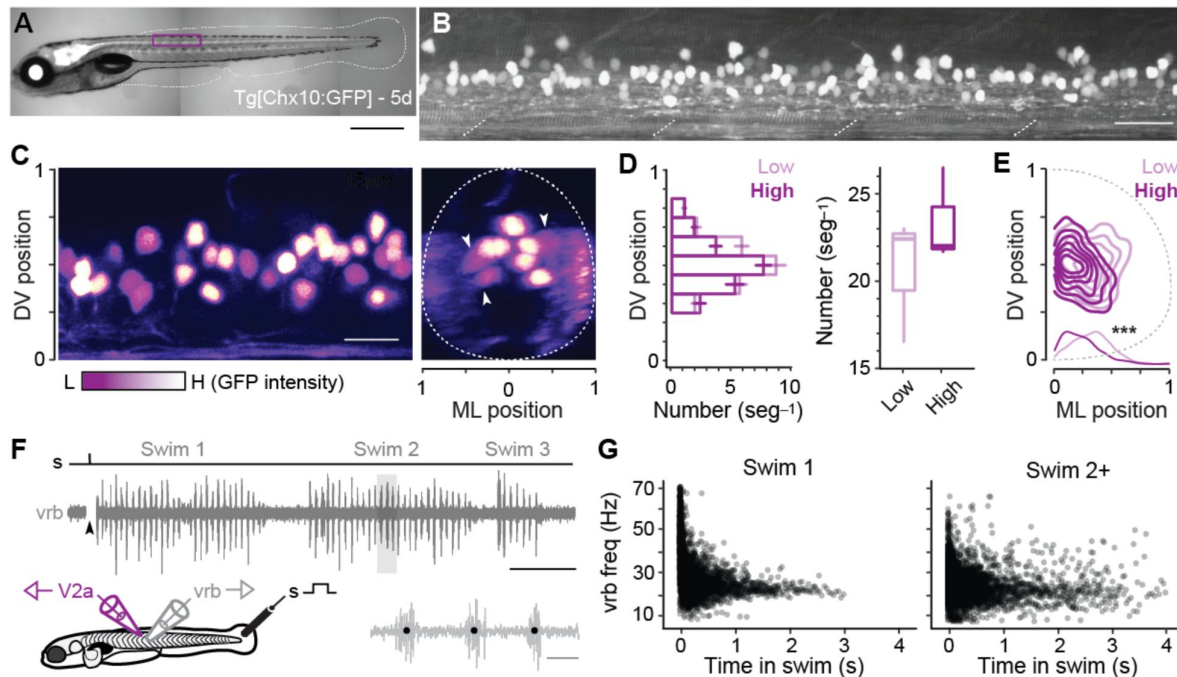


Figure 1

Identification of V2a neurons for recordings during 'fictive' swimming

- A)** Composite DIC/epifluorescence image of Tg[Chx10:GFP] larval zebrafish at 5 days post fertilization (5d). Scale bar, 0.5 mm.
- B)** Confocal image of the Tg[Chx10:GFP] spinal cord from segments 10-15 at midbody (purple box in A). Scale bar, 30 μ m.
- C)** Pseudo-colored images of V2a somata based on high (H, white) and low (L, purple) GFP intensity. Left panel: sagittal section, right panel: coronal section. Dashed lines indicate boundaries of spinal cord, which are normalized to 0-1 in the dorso-ventral and medio-lateral axes. White arrowheads indicate large, lateral V2a somata with low levels of GFP expression. Scale bar, 20 μ m.
- D)** Plots of numbers of low and high intensity V2a neurons per midbody segment ($n = 5-6$ segments from 3 fish). Left panel: Bar plot of average number ($\pm$ SEM) of low and high GFP V2a neurons along the dorsoventral (DV) axis. There is no significant difference in the medians of the dorsoventral distributions of high and low GFP expressing V2a neurons (Wilcoxon rank-sum test; $W = 68479$, $p = 0.6975$, $n = 347$ high and 401 low GFP neurons from 3 fish). Right panel: Box and whisker plot showing average number of high and low GFP neurons per segment (Wilcoxon rank-sum test, $W = 88$, $p = 0.05269$, $n = 17$ segments in 3 fish).
- E)** Two-dimensional contour density plots showing the mediolateral (ML) and dorsoventral (DV) distributions of V2a neurons expressing low or high levels of GFP. Density distributions of ML positions of V2a neurons are shown at the bottom of the contour plots, which are significantly different (***, Two-sample Kolmogorov-Smirnov test, $D = 0.16452$, $p = 8.46 \times 10^{-05}$, $n = 347$ high and 401 low GFP neurons from 3 fish).
- F)** Top panel: an example of an extracellular ventral rootlet recording of three consecutive swim episodes evoked by a mild electrical stimulus (s) to the tail fin (at arrowhead, artifact blanked). Scale bar, 500 ms. Bottom left panel: a cartoon of the locations of the stimulus and the recording electrodes. Bottom right panel: individual ventral rootlet bursts (vrb) on an expanded timescale. Black dots mark the center of each vrb used to calculate swim frequency in Hz (1/s), which is indicative of swimming speed. Scale bar, 20 ms.
- G)** Scatter plot of vrb frequency (Hz) versus the time in swim episode (ms). The swimming episode immediately following the stimulus is called Swim 1 (left), and any subsequent swimming episodes are called Swim 2+ (right).

Thus, different types of V2a neurons can be easily distinguished for targeting purposes and we can assess their firing and synaptic drive during fast anguilliform and slow carangiform modes distinguished by frequency and temporal sequence within fictive escape swim episodes. This set the stage for asking if the same mechanism of rhythmogenesis drives the phasic activation of distinct V2a types at different speeds.

Differences in intrinsic rhythmogenesis related to V2a neuron type

Rhythmogenesis can originate from cellular properties of synaptic drive or some combination of the two (Marder and Calabrese, 1996 [↗](#); Marder and Bucher, 2001 [↗](#)). In theory, neurons that are intrinsically rhythmogenic can create patterned outputs from unpatterned inputs. Thus, to explore the intrinsic rhythmicity of V2a neurons we performed whole-cell current-clamp recordings from high and low GFP-positive V2a neurons and tested their firing responses to stable current steps. By including dye in our recording pipette, we confirmed post-hoc that high GFP neurons were V2a-Ds and low GFP neurons were V2a-Bs (Fig. 2A [↗](#)). Differences in sizes and dorso-ventral distributions of V2a-D and V2a-B neurons in our current-clamp dataset ($n = 41$) were consistent with our analysis of high and low GFP neurons in stable lines (Fig. 2B [↗](#), left). Also, measurements of input resistance and rheobase current were consistent with differences in sizes and excitability between the two types (Fig. 2C [↗](#), left). Thus, we were confident of a sufficient sampling of V2a neurons to explore differences in rhythmogenic capacity related to type and recruitment order.

V2a-D neurons with ‘tonic’ responses and V2a-B neurons with higher frequency ‘chattering’ ones were distributed dorsally and had the lowest input resistances in their respective types (Fig. 2B-E [↗](#)). In tonic V2a-Ds, firing above rheobase current was observed consistently throughout the current step (Fig. 2E [↗](#), top), with instantaneous spike rates increasing linearly with increasing current, starting around 40 Hz and reaching 200 Hz (Fig. 2D [↗](#)). In chattering V2a-Bs, firing above rheobase current was intermittent throughout the step (Fig. 2F [↗](#), top), with instantaneous rates starting around 200 Hz and reaching 600 Hz (Fig. 2D [↗](#)). These observations are consistent with previous recordings from fast V2a-D and V2a-B neurons (Menelaou and McLean, 2019 [↗](#)).

On the other hand, V2a-D and V2a-B neurons with ‘burst’ responses were distributed ventrally and had the highest input resistances within their respective types (Fig. 2B-E [↗](#)). In bursting V2a neurons, firing above rheobase was characterized by higher frequency spikes superimposed on lower frequency membrane potential oscillations (Fig. 2E-F [↗](#), bottom). While the frequency of the slower oscillations increased with current (Fig. 2E-F [↗](#), bottom), instantaneous spike rates were already close to peak values at rheobase (500-600 Hz; Fig. 2D [↗](#)). Beyond 2-3x rheobase current, slower oscillations were no longer obvious and the firing pattern more closely resembled the chattering one (data not shown), as observed previously for motor neurons (Menelaou and McLean, 2012 [↗](#)).

Next, we compared the firing properties of distinct types of V2a neurons to their recruitment patterns during fictive escape swimming. As expected (Menelaou and McLean, 2019 [↗](#)), both tonic V2a-D and chattering V2a-B neurons were recruited at the beginning of the episode at median swim frequencies above 30 Hz (Fig. 2G-H [↗](#)), consistent with recruitment during fast anguilliform mode (Thorsen et al., 2004 [↗](#); McLean et al., 2008 [↗](#); McLean and Fetcho, 2009 [↗](#)). In contrast, burst V2a-Ds fired more reliably at the end of the episode at median frequencies that rarely exceeded 30 Hz (Fig. 2G-H [↗](#)), consistent with participation in slow carangiform mode (Thorsen et al., 2004 [↗](#); McLean et al., 2008 [↗](#); McLean and Fetcho, 2009 [↗](#)). Burst V2a-Bs were recruited just below 30 Hz, and fired more reliably throughout the episode and over a broader frequency range (Fig. 2G-H [↗](#)). A comparison between the step firing and swim firing patterns among V2a-D neurons revealed a remarkable correspondence (Fig. 2G [↗](#)), with median spike frequencies (tonic) or oscillation frequencies (burst) overlapping median recruitment frequencies during swimming (Fig. 2I [↗](#), left). However, the link between step firing and swim firing patterns was less obvious

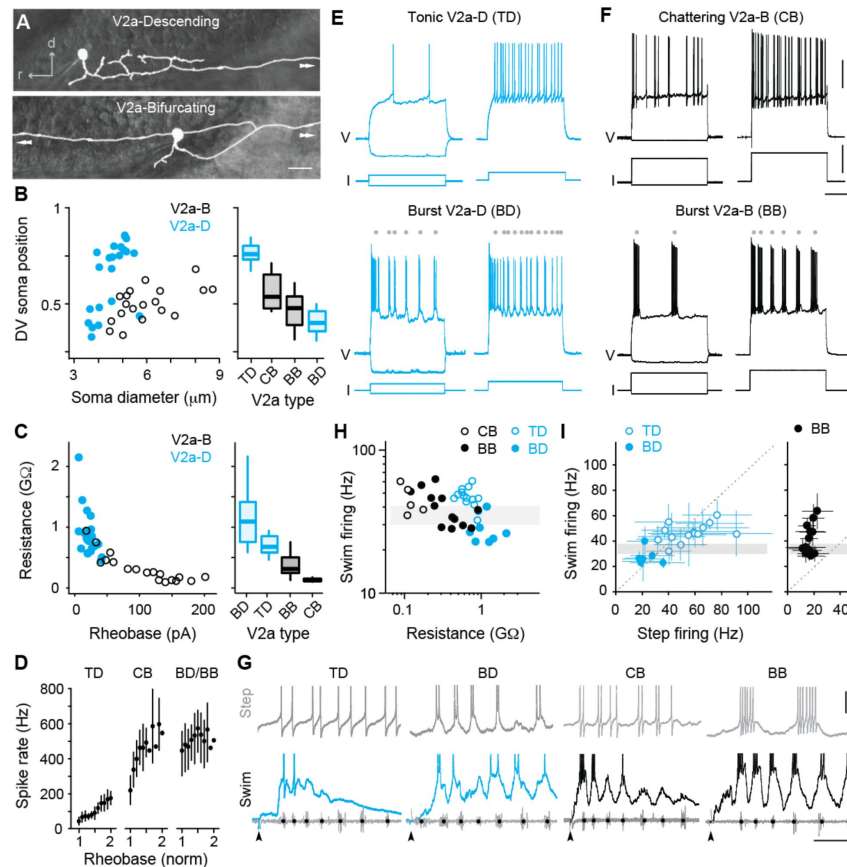


Figure 2

Cell-type-specific electrophysiological properties linked to recruitment order

A) Composite DIC/epifluorescent images of post-hoc fills of V2a-Descending (V2a-D, top) and V2a-Bifurcating (V2a-B, bottom) neurons. Arrows indicate continuation of axons outside the field of view. Scale bar, 10 μm .

B) Left panel: scatter plot of soma positions of V2a-D and V2a-B neurons normalized to dorsoventral (DV) position, with 1 demarcating the dorsal and 0 the ventral boundaries of the spinal cord. Right panel: boxplots of dorso-ventral soma positions of V2a-D and V2a-B neurons classified as tonic, bursting, or chattering based on their step-firing responses in panels *E* and *F*. TD=tonic V2a-D, CB=chattering V2a-B, BB=bursting V2a-B, BD=bursting V2a-D.

C) Left panel: scatter plot of input resistance ($\text{G}\Omega$) versus the rheobase (pA) of V2a-B (black) and V2a-D (blue) neurons. Right panel: Box plots of input resistance ($\text{G}\Omega$) of BD, TD, BB, and CB V2a neurons.

D) Mean ($\pm$ SD) instantaneous spike rates (Hz) between 1-2x rheobase for TD, CB, BD and BB neurons.

E) Top panel: step-firing responses of V2a-D neurons defined as tonic (TD). Bottom panel: step-firing responses of V2a-D neurons defined as bursting (BD). Grey dots depict the midpoint of slower membrane oscillations driving the bursting spiking behavior. Same scale as panel *F*.

F) Top panel: step-firing responses of V2a-B neurons defined as chattering (CB). Bottom panel: V2a-B neurons defined as bursting (BB). Grey dots depict the midpoint of slower membrane oscillations driving the bursting spiking behavior. Scale bars, 20 mV, 200 pA, 200 ms.

G) Top panels: Step-firing responses close to rheobase of TD, BD, CB, and BB neurons. Bottom panels: Swim firing responses evoked by a mild electrical shock shown on the same time scales. Black dots on ventral root bursts (grey) indicate swim frequency. Black arrows mark the stimulation artifact. Scale bars, 10 mV, 50 ms.

H) Scatter plot of median swim firing frequency (Hz) versus the input resistance ($\text{G}\Omega$) of CB, TD, BB, and BD on logarithmic x- and y-scales. Shaded grey box indicates 30-40 Hz range reflecting transition between anguilliform and carangiform swim modes.

I) A comparison of median swim-firing and median step-firing frequencies between 1-2x rheobase ($\pm$ SD) for TD, BD, and BB neurons. For TD neurons, step frequencies represent spike rates, while for BD and BB neurons they represent burst rates. CB neurons were too variable in step firing frequency to obtain reliable median values. Shaded grey box indicates 30-40 Hz range reflecting transition between anguilliform and carangiform swim modes.

for V2a-B neurons (**Fig. 2G**). For example, while bursting V2a-B neurons recruited at swim frequencies exceeding 60 Hz had higher frequency oscillations, step bursting rarely exceeded 20 Hz (**Fig. 2I**, right).

Thus, distinct electrophysiological properties define distinct V2a types recruited at different speeds and V2a-D neurons are better suited to providing intrinsic rhythmogenic activity during fast and slow swimming modes. This is consistent with their proposed rhythm-generating and pattern-forming roles (Menelaou and McLean, 2019), and suggests V2a-B neurons rely more on rhythmic synaptic drive to shape their phasic firing patterns during swimming.

Both types of V2a neuron operate at slow and fast speeds

Our current-clamp data suggest that different types would rely differently on cellular properties versus synaptic drive for rhythmogenesis. To test this idea more directly, we performed voltage-clamp recordings during fictive escape swimming. To more effectively isolate synaptic currents, we included cesium in the patch solution. Since this blocks voltage-gated channels and spiking, we needed to classify fast and slow V2a-D and V2a-B neurons subtypes by first monitoring spikes in cell-attached mode. Consistent with our current-clamp recordings, in our voltage-clamp dataset ($n = 39$) we found V2a-D and V2a-B neurons that fired at the beginning of the escape episode or nearer the end (**Fig. 3A**). V2a-D neurons fired more sparsely (**Fig. 3A**, left) and were divided into two distinct groups: 1) neurons that fired exclusively at the beginning of the episode at median swim frequencies exceeding 40 Hz, and 2) neurons that fired later on in the episode at median swim frequencies rarely exceeding 30 Hz (**Fig. 3B**, left; 3C). On the other hand, V2a-B neurons fired more reliably (**Fig. 3A**, right) and median values were more continuous, making it more difficult to group neurons based on location within the episode (**Fig. 3B**, right) or swim frequency (**Fig. 3D**).

To better segregate both types of V2a neurons based on speed, we used the frequency-dependent transition between anguilliform and carangiform modes to define V2a neurons recruited more reliably below 35 Hz as ‘slow’ and V2a neurons recruited more reliably above 35 Hz as ‘fast’ (**Fig. 4A**). Consistent with our current-clamp recordings, slow V2a-D and V2a-B neurons in our voltage-clamp recordings were more ventrally distributed with higher resistances than fast V2a-D and V2a-B neurons (**Fig. 4B**). A closer examination of spike timing relative to individual ventral rootlet bursts revealed median spike distributions in advance of the center of the motor burst, as expected for premotor excitatory interneurons involved in rhythmogenesis (**Fig. 4C**). Plots of spike densities normalized to center burst revealed that V2a-B neurons begin to fire slightly in advance of V2a-D neurons (**Fig. 4D**), consistent with differences in membrane resistance related to size, where neurons with faster time constants respond more quickly to synchronous inputs (Menelaou and McLean, 2019).

Next, to explore recruitment probability as a function of speed in more depth we divided the full range of fictive swimming frequencies into four bins distributed around 35 Hz. For V2a-D neurons, fast neurons were rarely recruited below 35 Hz and increased to a maximum of 50% recruitment reliability per motor burst above 35 Hz (**Fig. 4E**, top left). Similarly, slow V2a-D neurons did not exceed 50% recruitment reliability per motor burst below 35 Hz and rarely fired above it (**Fig. 4E**, top right). When collapsed into two bins below and above 35 Hz, these differences in recruitment probability were significant (**Fig. 4F**, top). For V2a-B neurons, fast neurons fired more reliably below 35 Hz than fast V2a-Ds and increased to 100% recruitment reliability per motor burst above 35 Hz (**Fig. 4E**, bottom left). Similarly, slow V2a-B neurons fired more reliably above 35 Hz than slow V2a-Ds, and at 100% reliability below it (**Fig. 4E**, bottom right). Binning of recruitment among V2a-Bs revealed significant differences related to speed among the fast subtypes only (**Fig. 4F**, bottom). Among the slow V2a-Bs, decreases in recruitment reliability were observed in the majority of neurons ($n = 5$ of 6), however the difference was not significant when pooled (Wilcoxon signed-rank test, $V=20$, $p=0.0625$, $n = 6$).

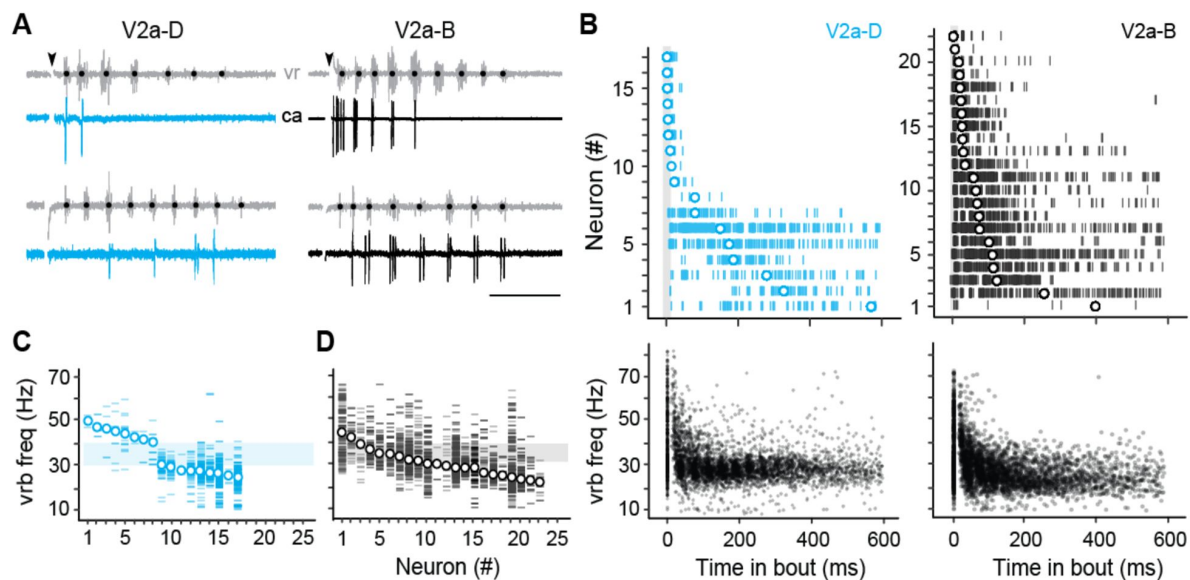


Figure 3

Differences in recruitment patterns among the distinct V2a types

A) Left panel: Cell-attached (ca) recordings of descending V2a (V2a-D) neurons during fictive swimming evoked by a brief stimulus (at black arrowhead). Black dots on ventral root (vr) bursts (grey) indicate swim frequency. The top neuron fires immediately after the stimulus during high frequency swimming, while the bottom neuron fires near the end of the episode at lower frequencies. Right panel: As shown to the left, but for bifurcating V2a (V2a-B) neurons. Scale bar, 100 ms.

B) Top panel: spike timing of V2a-D (left) and V2a-B (right) neurons relative to the start of swimming ordered by the median of the distribution. Bottom panel: A scatterplot of vrb frequency (Hz) as a function of time for all episodes (Swim 1 and 2+) during which the V2a-D (left) or V2a-B (right) neurons were recruited.

C) Raster plots of ventral root burst frequencies (Hz) over which V2a-Ds were recruited ordered by the median vrb recruitment frequency (Hz). Shaded box indicates 30-40 Hz range reflecting transition between anguilliform and carangiform swim modes.

D) Raster plots of ventral root burst frequencies (Hz) over which V2a-Bs were recruited ordered by the median vrb recruitment frequency (Hz). Shaded box indicates 30-40 Hz range reflecting transition between anguilliform and carangiform swim modes.

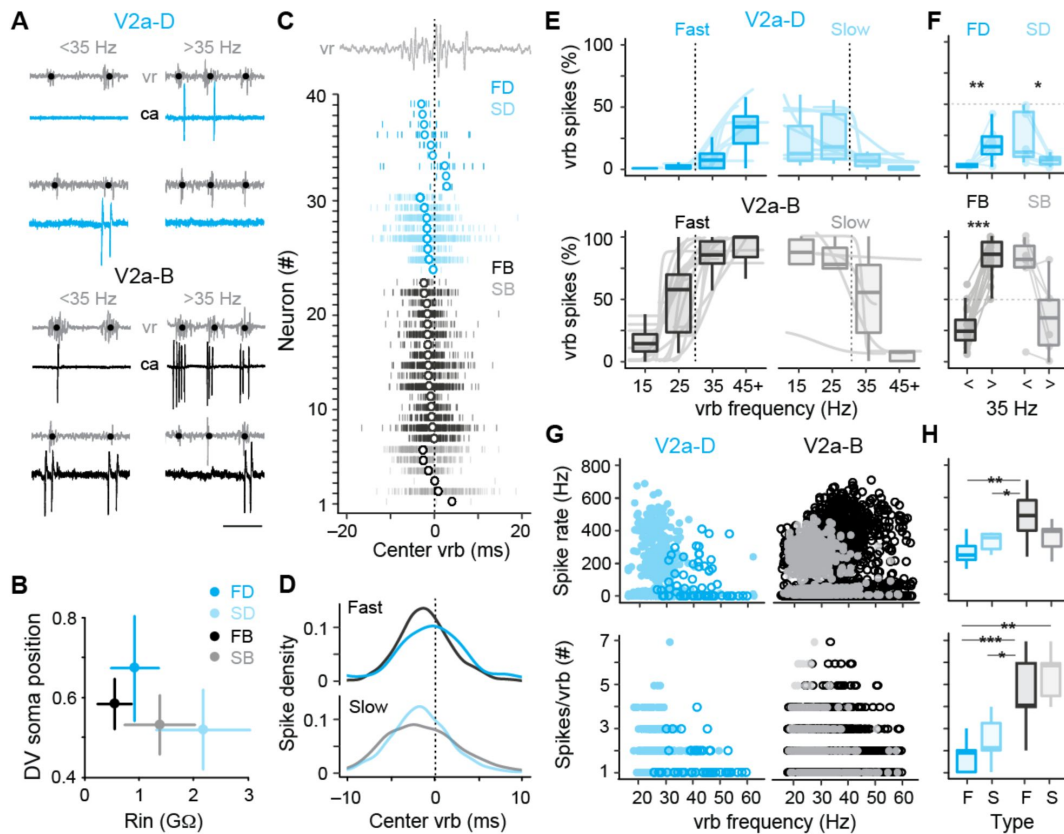


Figure 4

Distinct types of slow and fast V2a neurons distinguished based on recruitment

A) Top panel: Cell-attached (ca) and ventral rootlet (vr) recordings during slow (<35 Hz) and fast (>35 Hz) swimming illustrate V2a-D neurons (blue) that fire exclusively at slow and fast speeds. Bottom panel: V2a-B neurons (black) fire more reliably at both slow and fast speeds. Scale bar, 25 ms.

B) Plots of the mean ($\pm$ SD) dorsoventral (DV) positions of fast (FD) and slow (SD) V2a-Ds, and fast (FB) and slow (SB) V2a-Bs versus their mean ($\pm$ SD) input resistance (Rin).

C) Spike timing of slow (S) and fast (F) V2a-D and V2a-B neurons relative to the center of the ventral root burst (vr) depicted by the dashed vertical line. Tick marks represent individual spikes from multiple cycles and circles represent median spike timing.

D) Density plots of spike timing relative to the center burst for slow and fast V2a-B and V2a-D neurons illustrated in panel C.

E) Top panel: Box plots of the percentage of ventral root bursts with spikes (vrb spikes %) as a function of vrb frequency (Hz) for fast (FD, $n = 9$) and slow (SD, $n = 8$) V2a-D neurons. For box plots, the ventral root burst frequency was binned at 15 Hz intervals starting at 15 Hz, and the final interval of 45+Hz consisted of a 20 Hz range spanning from 45-65 Hz. Box plots are superimposed on trendlines from individual neurons fit to data binned at 5 Hz intervals, whose slopes define whether they are fast (positive) or slow (negative). Dashed line indicates transition between slow carangiform and fast anguilliform modes. Bottom panel: As above but for fast (FB, $n = 16$) and slow (SB, $n = 6$) V2a-B neurons.

F) Top panel: Box plots of vrb spikes % for slow and fast V2a-D neurons from panel E collapsed into single slow (<35 Hz) and (>35 Hz) bins for purposes of statistical analysis (Wilcoxon signed-rank test; FD, $V = 1$, $p = 0.007812$, $n = 9$; SD, $V = 36$, $p = 0.007813$, $n = 8$). Bottom panel: As above but for V2a-B neurons (Wilcoxon signed-rank test: FB, $V = 0$, $p = 6.104e-05$, $n = 16$; SB, $V = 20$, $p = 0.0625$, $n = 6$).

G) Top panel: Scatter plots of instantaneous spike rates versus vrb frequency (Hz) for V2a-D (left) and V2a-B (right) neurons color coded as fast (dark) and slow (light). FD, $n = 9$; SD, $n = 8$; FB, $n = 16$; SB, $n = 6$. Bottom panel: Scatter plots of the number of spikes per ventral root burst (#) as a function of vrb frequency (Hz), organized and color coded as above.

H) Top panel: Box plots of maximum instantaneous spike rates for fast ($n = 5$) and slow ($n = 7$) V2a-D and fast ($n = 16$) and slow ($n = 5$) V2a-B neurons. Note, only a subset of V2a neurons that fired two or more spikes per cycle were included in this analysis (Wilcoxon rank-sum test; FB-FD, $W = 73$, $p = 0.00412$; FB-SD, $W = 90$, $p = 0.02247$). Bottom panel: Box plots of the maximum number of spikes/vrb (Wilcoxon rank-sum test; FB-FD, $W = 132.5$, $p = 0.0005348$; FB-SD, $W = 100.5$, $p = 0.024$; FD-SB, $W = 50$, $p = 0.00667$; FD, $n = 9$; SD, $n = 8$; FB, $n = 16$; SB, $n = 6$).

In further support of the speed-based modular organization of V2a neurons, in slow types higher instantaneous spike rates were observed at lower burst frequencies, while in fast types higher spike rates were observed at higher burst frequencies (**Fig. 4G** [↗](#), *top*). Critically, spike rates observed during fictive swimming overlap with those observed during current steps between 1-2x rheobase, ranging from in 200-600 Hz (**Fig. 4H** [↗](#), *top*), with the fastest spike rates observed in fast V2a-B neurons. This suggests that the firing patterns observed during current injections between 1-2x rheobase are functionally relevant. When we measured spike numbers per burst, lower numbers were consistently observed at higher burst frequencies regardless of type or speed class (**Fig. 4G** [↗](#), *bottom*) consistent with less time to spike during shorter cycle periods. In addition, V2a-B neurons fired significantly more spikes per burst than V2a-Ds on average (**Fig. 4H** [↗](#), *bottom*), consistent with their higher recruitment probabilities.

Thus, within our voltage-clamp dataset we have V2a-D and V2a-B neurons operating during anguilliform and carangiform swimming modes. While V2a-D neurons fire more sparsely and in a stricter modular fashion, V2a-B neurons fire more reliably and over a broader range of frequencies, consistent with occupation of a pattern forming layer downstream of distinct fast and slow rhythmic signals.

Differences in the levels of synaptic drive mirror differences in speed modularity

To get a better sense of the relative segregation of synaptic inputs to V2a-D and V2a-B neurons based on speed, we first focused on the maximum amplitude of excitatory and inhibitory currents recorded during fictive escape swimming. Clamping V2a neurons at a potential to isolate excitatory post-synaptic currents (EPSCs) revealed faster time course, compound EPSCs superimposed on a slower, tonic inward current (**Fig. 5A,B** [↗](#)). While tonic excitation was observed the entirety of the episode, fast V2a-Ds received compound EPSCs exclusively during high frequency swimming at the beginning (**Fig. 5A** [↗](#), *top*), while slow V2a-Ds received compound EPSCs exclusively during slower swimming later in the episode (**Fig. 5A** [↗](#), *bottom*), consistent with their recruitment patterns. A similar pattern was observed among V2a-Bs, however compound EPSCs were observed over a broader range of frequencies (**Fig. 5B** [↗](#)), also consistent with their recruitment patterns. We also noted differences in maximum levels of current related to type. When we plotted maximum current (tonic current + compound EPSCs) versus input resistance, we observed a significant negative correlation, with lower resistance V2a-Bs receiving orders of magnitude more excitation than higher resistance V2a-Ds (**Fig. 5C** [↗](#)). This is consistent with peak excitatory drive normalized to cellular excitability, as reported previously (Menelaou et al., 2022 [↗](#)).

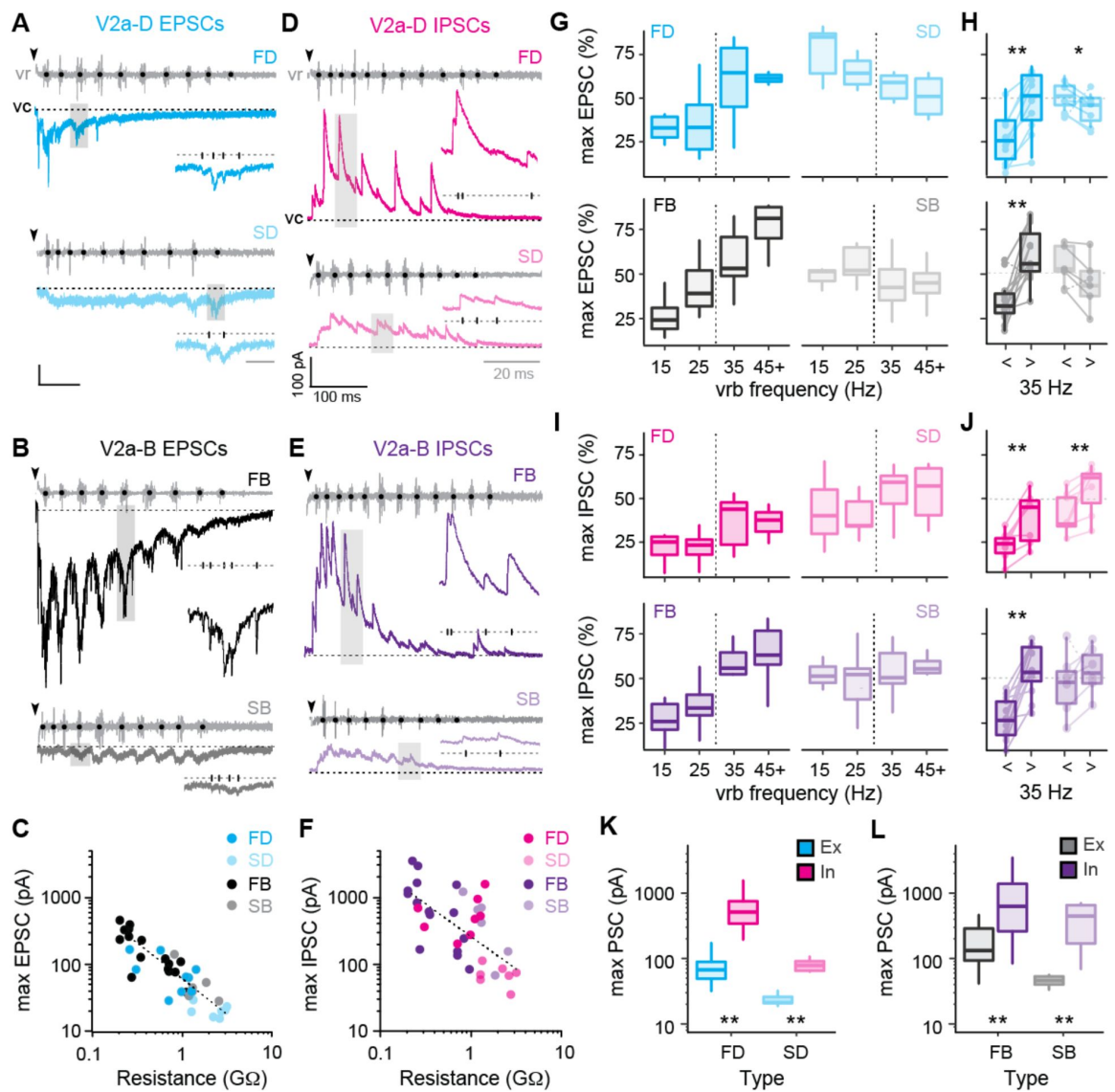


Figure 5

Differences in levels of synaptic drive related to speed and V2a cell type

A) Voltage-clamp (vc) recordings of excitatory post-synaptic currents (EPSCs) received by fast (top, FD) and slow (bottom, SD) V2a-D neurons along with ventral root (vr) recordings during fictive swimming triggered by an electrical stimulus (black arrowheads, artifact blanked). Black dots indicate burst intervals for frequency measures. Grey shaded boxes indicate region expanded inset, illustrating the holding potential (dashed line, -75 mV) and individual EPSCs (tick marks). Scale bars, 25 pA, 100 ms (20 ms inset).

B) As in A, but for fast (top, FB) and slow (bottom, SB) V2a-B neurons.

C) Scatter plot of maximum EPSC amplitude (pA) as a function of input resistance ($G\Omega$) on logarithmic x- and y-scales. Dashed logarithmic trendlines are included for illustrative purposes (Spearman's rank correlation test, $Rho = -0.8730769$, $p < 2.2 \times 10^{-13}$, $n = 39$).

D) Voltage-clamp recordings of inhibitory post-synaptic currents (IPSCs) at a holding potential of 10 mV from fast (top, FD) and slow (bottom, SD) V2a-D neurons, organized as detailed in panel A. Scale bars, 100 pA, 100 ms (20 ms inset).

E) As in D, but fast (top, FB) and slow (bottom, SB) V2a-B neurons.

F) Scatter plot of maximum IPSC amplitude as a function of input resistance ($G\Omega$) on logarithmic x- and y-scales (Spearman's rank correlation test; $Rho = -0.6332046$, $p = 5.041 \times 10^{-05}$, $n = 36$).

G) Top panel: Box plots of the maximum EPSC per cycle as a percentage of the maximum current (max EPSC%) for fast (FD, $n = 9$) and slow (SD, $n = 6$) V2a-D neurons, organized into 4 bins based on frequency. Dashed line indicates transition between slow carangiform and fast anguilliform modes. Bottom panel: as above, but for fast (FB, $n = 11$) and slow (SB, $n = 5$) V2a-B neurons.

H) Top panel: Box plots of max EPSC% for fast and slow V2a-D neurons from panel G, collapsed into single slow (<35 Hz) and fast (>35 Hz) bins for purposes of statistical analysis (Wilcoxon signed-rank test; FD, $V = 0$, $p = 0.003906$, $n = 9$; SD, $V = 21$, $p = 0.03125$, $n = 6$). Bottom panel: As above but for V2a-B neurons (Wilcoxon signed-rank test; FB, $V = 2$, $p = 0.001953$, $n = 11$; SB, $V = 15$, $p = 0.0625$, $n = 5$). Boxed plots are superimposed on trendlines from individual neurons. Note, 1 cell each for V2a-D and V2a-B slow subtypes behaved more like fast subtypes (dashed lines).

I) Top panel: Box plots of the maximum IPSC per cycle as a percentage of the maximum current (max IPSC%) for fast (FD, $n = 7$) and slow (SD, $n = 7$) V2a-Ds organized as in panel G. Bottom panel: As above but for fast (FB, $n = 12$) and slow (SB, $n = 6$) V2a-B neurons.

J) Top panel: Box plots of max IPSC% for fast and slow V2a-D neurons from panel I, collapsed into single slow (<35 Hz) and fast (>35 Hz) bins for purposes of for statistical analysis (Wilcoxon signed-rank test; FD, $V = 0$, $p = 0.01563$, $n = 7$; SD, $V = 1$, $p = 0.03125$, $n = 7$). Bottom panel: as above, but for fast and slow V2a-B neurons (Wilcoxon signed-rank test; FB, $V = 0$, $p = 0.0004883$, $n = 12$; SB, $V = 4$, $p = 0.2188$, $n = 6$).

K) Box plots of maximum excitatory and inhibitory current for FD (Wilcoxon signed-rank test; $V = 0$, $p = 0.00781$, $n = 9$ for excitation, $n = 8$ for inhibition) and SD (Wilcoxon signed-rank test; $V = 0$, $p = 0.01563$, $n = 8$ for excitation, $n = 7$ for inhibition) neurons.

L) As in K but for FB ($V = 1$, $p = 0.0001221$, $n = 16$ for excitation, $n = 15$ for inhibition) and SB ($V = 0$, $p = 0.03125$, $n = 6$ for excitation, $n = 6$ for inhibition) neurons.

Next, clamping V2a neurons at a potential to isolate inhibitory post-synaptic currents (IPSCs) revealed faster time course compound IPSCs superimposed on a slower, tonic outward current (**Fig. 5D-E**). In all V2a types tonic inhibition persisted the entirety of the episode and compound IPSCs were larger at higher frequencies near the beginning (**Fig. 5D-E**). Like excitation, plots of maximum current (tonic current + compound IPSCs) revealed a significant negative correlation between inhibitory current and input resistance (**Fig. 5F**), meaning that peak inhibitory drive is also normalized to cellular excitability. Given these large differences in maximum current values, to compare frequency-dependent changes in excitation and inhibition, we normalized the currents for each cycle to the maximum currents for the neuron (**Fig. 5C**, 5F). So, maximum EPSCs or IPSCs in a given cycle are expressed as a percentage of the maximum value in all cycles at any speed.

For fast V2a-D and V2a-B neurons, maximum excitatory drive increased with increasing frequency (**Fig. 5G**, left). Differences in maximum excitation were significant for both types when data were pooled (**Fig. 5H**), consistent with their recruitment patterns (**Fig. 4F**). Inhibition also followed the same pattern for fast V2a neurons (**Fig. 5I**, left), significantly increasing at fast speeds (**Fig. 5J**). For slow V2a-D neurons excitation was significantly lower at higher frequencies (**Fig. 5G**, top right; 5H, top) and inhibition was significantly higher (**Fig. 5I**, top right; 5J, top). However, for slow V2a-B neurons neither excitation (**Fig. 5G**, bottom right; 5H, bottom) nor inhibition (**Fig. 5I**, bottom right; 5J, bottom) changed significantly when pooled, despite the decreases in excitation ($n = 4$ of 6) and increases in inhibition ($n = 5$ of 6) in the majority of neurons. These patterns of excitation and inhibition are consistent with recruitment patterns, where slow V2a-D neurons are silenced at fast speeds and slow V2a-B neurons are recruited albeit with lower reliability (**Fig. 4F**).

While the increase in inhibition at fast speeds to slow V2a neurons makes sense in terms of their disengagement during fast anguilliform mode (McLean et al., 2008), the increase in inhibition to fast V2a neurons clearly does not have a similar impact, despite being normalized to excitability. Indeed, when we compared the maximum excitation and inhibition to V2a-Ds (**Fig. 5K**) and V2a-Bs (**Fig. 5L**), maximum inhibition was orders of magnitude larger than maximum excitation. Even when controlling for differences in driving forces experienced at resting potentials, it would be difficult for V2a neurons to fire if excitation and inhibition arrived concurrently.

Thus, measurements of maximum levels of excitatory and inhibitory drive that include fast phasic and slower tonic components are consistent with recruitment patterns. These data also suggest the timing of inhibition plays a critical role in shaping the rhythmic output of V2a neurons more broadly.

Differences in the timing of synaptic drive better explain rhythmic V2a recruitment

To compare the timing of compound EPSCs and IPSCs we normalized the amplitude to the maximum (1) and minimum (0) values for that cycle and normalized the timing to the phase of the swim cycle, with the centers of two successive motor bursts demarcating phase 0 and 1 (**Fig. 6A**). To simplify comparisons between peaks in excitation and troughs in inhibition, we inverted the sign of excitation. By convention, synaptic currents arriving between 0.75 – 0.25 are described as ‘in-phase’ with local motor output, while currents arriving 0.25 – 0.75 are described as ‘anti-phase’. In fish, in-phase currents are assumed to arise from recurrent sources on the ipsilateral side, while anti-phase currents arise from the reciprocal sources on the contralateral side. This means that differences in the timing of peaks of excitation and inhibition can be used to draw inferences about patterns of connectivity along and across the body.

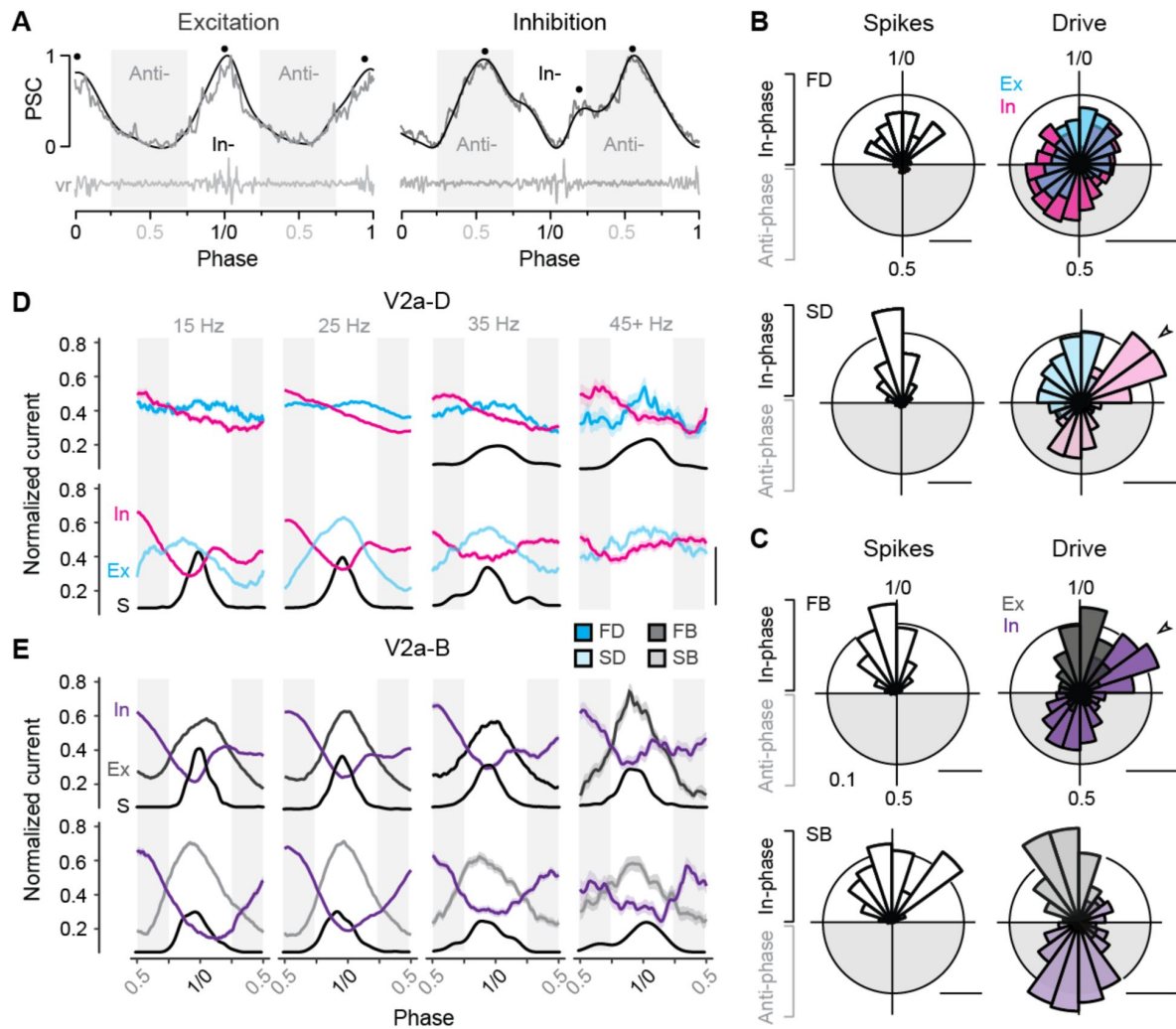


Figure 6

Differences in timing of synaptic drive related to speed and V2a cell type

A Raw (grey) and low-pass filtered (black) traces of post-synaptic current (PSC), normalized to the peak (1) and trough (0) of phasic excitation (left) and inhibition (right). Corresponding ventral root (vr) bursts used to define in- and anti-phase (gray shaded boxes) currents are shown below. Block dots denote the peaks identified by the findpeaks() function in MATLAB. Note, excitation has been inverted to simplify comparisons to inhibition.

B Left panel: Circular bar plots of spike phase for fast (FD, $n = 9$) and slow (SD, $n = 8$) V2a-D neurons. Right panel: Circular bar plots of phase normalized peaks in low pass filtered EPSCs (FD, $n = 9$; SD, $n = 7$) and IPSCs (FD, $n = 7$; SD, $n = 7$). Arrowhead indicates prominent in-phase IPSC component for slow V2a-Ds. Scale bars, 10% total distribution.

C Left panel: Circular bar plots of spike phase for fast (FB, $n = 16$) and slow (SB, $n = 6$) V2a-B neurons. Right panel: Circular bar plots of phase normalized peaks in low pass filtered EPSCs (FB, $n = 11$; SB, $n = 6$) and IPSCs (FB, $n = 12$; SB, $n = 6$). Arrowhead indicates prominent in-phase IPSC component for fast V2a-bs. Scale bars, 10% total distribution.

D Line plots from V2a-D neurons of averaged excitation (Ex; FD, $n = 9$; SD, $n = 6$), averaged inhibition (In; FD, $n = 7$; SD, $n = 7$), and spike densities (S; FD, $n = 7$; SD, $n = 7$) broken into four ventral root burst frequency (Hz) bins. Post synaptic current was normalized on a cycle-by-cycle basis, such that the minimum and maximum current per cycle equaled 0 and 1 respectively. Scale bar for spike densities, 50% total distribution.

E As in panel D, but averaged excitation (Ex; FB, $n = 11$; SB, $n = 5$), inhibition (In; FB, $n = 12$; SB, $n = 6$), and spike densities (S; FB, $n = 16$; SB, $n = 6$) for V2a-B neurons.

First, to assess differences in the timing of compound EPSCs and IPSCs, we low pass filtered the raw current trace to identify peaks in-phase and anti-phase (**Fig. 6A** [↗](#), *black dots*), which were then plotted on a circular axis and compared to cell-attached spiking activity (**Fig. 6B-C** [↗](#)). For all V2a neurons, spikes were exclusively distributed in-phase with local motor output (**Fig. 6B** [↗](#), *left*; **6C**, *left*), however there were type-specific differences in the distributions of peak excitation and inhibition.

For fast V2a-Ds phasic excitation began early in anti-phase, with a peak distribution in-phase coinciding with a dip in phasic inhibition and spiking activity (**Fig. 6B** [↗](#), *top*). Phasic inhibition was also broadly distributed, with peak values in anti-phase (**Fig. 6B** [↗](#), *top right*). For slow V2a-Ds, phasic excitation was also phase advanced relative to spiking activity (**Fig. 6B** [↗](#), *bottom*), with peaks obvious earlier in anti-phase (**Fig. 6B** [↗](#), *bottom right*). Unlike fast V2a-Ds, however, there were two clear peaks in phasic inhibition, one immediately trailing in-phase excitation (**Fig. 6B** [↗](#), *bottom right arrow*) and one peaking anti-phase. Notably, the presence of in-phase inhibition was also associated with a more coherent distribution of spikes in the slow V2a-Ds (c.f., **Fig. 6B** [↗](#), *left*). For fast and slow V2a-Bs, phasic excitation was found exclusively in-phase, coincided well with spike distributions and was more coherent than V2a-Ds (**Fig. 6C** [↗](#)), consistent with the idea they are downstream of V2a-D neurons. Remarkably, however, the pattern of phasic inhibition was inverted, such that fast V2a-Bs received peaks in-phase and anti-phase (**Fig. 6C** [↗](#), *top right arrow*), while slow V2a-Bs received predominantly anti-phase inhibition (**Fig. 6C** [↗](#), *bottom right*). Differences in the predominance of in-phase inhibition between V2a-B types were also associated with differences in the coherence of spiking activity (**Fig. 6C** [↗](#), *left*), as observed for V2a-D types (**Fig. 6B** [↗](#), *left*).

Next, to get a better sense of how patterns of phasic excitation and inhibition differ within types at different speeds, we binned the data by frequency as above and examined averages of normalized raw current traces. Among the V2a-Ds, fast subtypes exhibited reciprocal inhibition at all frequencies, with phasic peaks in excitation and inhibition most obvious at higher frequencies coincident with spiking activity (**Fig. 6D** [↗](#), *top*). Among slow V2a-Ds, coherent peaks in reciprocal and recurrent inhibition were most obvious at slow speeds coincident with phasic excitation and spiking behavior (**Fig. 6D** [↗](#), *bottom*). Among the V2a-Bs, patterns of reciprocal and recurrent inhibition were consistent across all speed bins, with peaks in phasic excitation and dips in phasic inhibition coincident with spiking activity (**Fig. 6E** [↗](#)).

Thus, despite differences in cellular rhythmogenic capacity, these observations suggest all types of V2a neurons rely on phasic excitation and inhibition to sculpt their rhythmic firing behavior during swimming. Peaks in phasic excitation arrive earlier among V2a-D neurons, consistent with their hierarchical positioning upstream from V2a-B neurons, and the relative coherence of phasic excitatory and inhibitory drive at different frequencies is consistent with speed modularity in this population. Among V2a-B neurons, the coherence of phasic excitatory and inhibitory drive over the entire frequency range is consistent with a role integrating inputs from the different speed modules. Critically, these data also suggest reciprocal inhibition plays a key role in rhythmogenesis at fast speeds, while recurrent inhibition plays a more dominant role in rhythmogenesis at slow speeds.

Discussion

Our work was motivated by the opaque origins of spinal rhythmicity at different locomotor speeds and the potential for zebrafish to clarify this fundamental issue. By recording from different subtypes of V2a neurons during slow and fast fictive swimming we have, for the first time, linked cellular and synaptic properties to recruitment patterns over nearly a full operational range of locomotion. First, our recordings reveal that all V2a neurons demonstrate some form of rhythmicity in response to current steps. The idea that different types of V2a interneurons exhibit

different firing properties is not unexpected (Dougherty and Kiehn, 2010 [DOI](#); Zhong et al., 2010 [DOI](#); Menelaou and McLean, 2019 [DOI](#); Song et al., 2020 [DOI](#)). However, thanks to our ability to link cellular properties to natural recruitment patterns we reveal V2a-Ds alone would qualify as ‘pacemakers’ (Marder and Calabrese, 1996 [DOI](#); Marder and Bucher, 2001 [DOI](#)), with speed-dependent differences in intrinsic firing mode within V2a-Ds. Specifically, V2a-D neurons recruited at swim frequencies below ~40 Hz during carangiform swimming generate intrinsic bursting at those frequencies, while V2a-D neurons recruited at frequencies above ~40 Hz during anguilliform swimming generate tonic firing at those frequencies. In V2a-B neurons, chattering and bursting properties in response to current steps do not reflect their natural firing patterns during swimming, and so we expected they would rely more on pacemaking inputs from V2a-Ds to shape their rhythmicity.

Surprisingly, however, we found that both V2a-Ds and V2a-Bs receive phasic excitatory input, and so synaptic signals are already rhythmic by the time they arrive. This could be because our recordings were performed at midbody and so rhythms are generated by more rostral V2a-Ds with identical cellular properties. Optogenetic and lesion studies support a rostral bias in spinal rhythmicity in larval zebrafish (Wiggin et al., 2012 [DOI](#); Kimura et al., 2013 [DOI](#)), consistent with studies of spinal locomotor networks in lampreys and tadpoles, where neurons nearer the hindbrain/spinal cord boundary have a higher capacity for rhythmogenesis (Li et al., 2010 [DOI](#); Buchanan, 2018 [DOI](#)). On the other hand, bursting properties could serve as a source of amplification for synaptic inputs to ensure reliable recruitment of rhythmogenic neurons or to ensure rhythmogenesis in cases where synaptic drive fails or the spinal cord is damaged (Messina et al., 2017 [DOI](#); Parker, 2017 [DOI](#); Cangiano and Grillner, 2018 [DOI](#); McClellan, 2018 [DOI](#)).

Another unexpected observation is that rhythmogenesis in V2a-D neurons during slow, carangiform mode relies on recurrent inhibition. Previous lesion and ablation studies in adult lampreys (Buchanan and McPherson, 1995 [DOI](#)) and embryonic frogs (Moult et al., 2013 [DOI](#)) suggest reciprocal inhibition is essential for rhythmogenesis, consistent with the half-center model. Our recordings from fast V2a-Ds are also consistent with the importance of reciprocal inhibition in rhythmogenesis, however this is only during anguilliform mode. Critically, lampreys and frog embryos use anguilliform swimming at all speeds (Hoff and Wassersug, 2000 [DOI](#); McClellan et al., 2016 [DOI](#)), suggesting that this mode of swimming is more reliant on bilateral coordination to sustain spinal rhythms.

On the other hand, the speed-dependent dominance of reciprocal versus recurrent inhibition is flipped when considering pattern-forming V2a-B neurons. Until now recurrent inhibition was linked to limiting motor output (Hultborn et al., 1979 [DOI](#)), not to rhythmogenesis (see, however, Linden and Berg, 2021 [DOI](#)). The stronger recurrent inhibition we see in fast V2a-Bs during anguilliform swimming is consistent with this role. As swimming speed increases, motor bursts must occupy a consistent fraction of the swim cycle to ensure propulsive thrust and avoid co-incident bilateral activity (Grillner, 1974 [DOI](#)). Fast V2a-Bs have relatively strong connections to fast motor neurons (Menelaou and McLean, 2019 [DOI](#)), so limiting their firing would ensure that as swim frequency increases and cycle periods decrease, motor bursts decrease.

Among slow V2a-Bs recurrent inhibition is largely absent in favor of reciprocal inhibition. Assuming they follow the same connectivity rules as fast V2a-B types, this means at lower frequencies there is a greater capacity for longer motor bursts via a broader window of premotor excitation. This supports our recent study exploring the contribution of V2a-D and V2a-B neurons to slow biased swimming during optomotor responses (Jay et al., 2023 [DOI](#)). Optomotor responses in larval zebrafish involve increases in the duration of motor output on one side via unilateral recruitment of motor neurons, without impacting the timing of left-right or head-tail coordination. While V2a-D neurons are agnostic to turn direction, V2a-Bs are selectively recruited on the turning side. This provides a mechanism to decouple swim intensity and frequency, as required for direction control during locomotion.

In adult zebrafish, V2a and motor neurons have been grouped into slow, intermediate, fast and escape modules based on patterns of recruitment and phasic synaptic drive during fictive swimming (El Manira, 2023 [↗](#)). In larvae, we also observe recruitment patterns among V2a neurons that would qualify as slow, intermediate, fast and escape (c.f., **Fig. 2G** [↗](#)). However, we also find that recruitment probability varies based on morphological type, with V2a-D neurons always firing more sparsely than V2a-B neurons, regardless of speed. Collectively, these observations suggest that increases in speed in adults and larvae are initially accomplished via more reliable V2a neuron recruitment, as reported in mice (Zhong et al., 2011 [↗](#)). This fits studies of motor neurons, where relatively low initial forces rely on recruitment (Enoka and Duchateau, 2017 [↗](#); Jay et al., 2023 [↗](#)), while higher forces are achieved by increasing firing rates (Bhatt et al., 2007 [↗](#); Enoka and Duchateau, 2017 [↗](#)), in addition to silencing more easily recruited neurons (Smith et al., 1980 [↗](#); Kyriakatos et al., 2011 [↗](#); Menelaou and McLean, 2012 [↗](#)).

In sum, our findings suggest that cell-type-specific modes of spinal rhythmogenesis are responsible for different speeds of locomotion. Studies in mice have implicated En1-labeled V1 interneurons not only in reciprocal inhibition (Zhang et al., 2014 [↗](#)), but also in recurrent inhibition, perhaps most famously via Renshaw cells (Alvarez et al., 2005 [↗](#)). The hierarchical and modular nature of recurrent inhibition we find here could explain the relative challenge disentangling the role of V1 neurons in rhythm-generation versus pattern-formation in both zebrafish (Kimura and Higashijima, 2019 [↗](#)) and mice (Gosgnach et al., 2006 [↗](#)). Ultimately, these observations suggest that a common mechanism of rhythmogenesis is insufficient to explain the full range operational range of locomotion.

Materials and methods

Fish husbandry

Tg[Chx10:GFP] fish (ZFIN ID: ZDB-ALT-061204-2) were used to target V2a neurons in the spinal cord (Kimura et al., 2006 [↗](#)). Fish were maintained at 28.5°C on a 14 hr light/10 hr dark cycle in an aquatics facility staffed by Northwestern University's Center for Comparative Medicine. All experiments were performed at room temperature between 4 and 5 days post fertilization (dpf) when zebrafish larvae are free swimming. Because 4-5 dpf larvae have not yet developed secondary sexual characteristics, their sex cannot be easily determined. Therefore, experiments were performed on larvae of either sex. All experimental protocols conform to National Institutes of Health guidelines on animal experimentation and were approved by the Northwestern University Institutional Animal Care and Use Committee (IS00019359 and IS00019319).

Electrophysiology

Electrophysiological recordings were acquired using a Multiclamp 700A amplifier (Molecular Devices), a Digidata series 1322A digitizer (Molecular Devices), and pClamp 10 software (Molecular Devices). The intrinsic firing patterns of V2a neurons were assessed by performing 500 ms long current steps at 1-2x rheobase in current-clamp mode. To compare the intrinsic firing properties of V2a neurons to their recruitment profiles during 'fictive' escape swimming, we performed simultaneous whole cell current-clamp recordings of V2a neurons and extracellular recordings of the peripheral ventral rootlet. Synaptic currents during fictive swimming were assessed in dual whole voltage-clamp and extracellular rootlet recordings. Swimming was evoked by applying a mild and brief electrical shock (2-10 V; 0.1-0.4 ms) using a bipolar tungsten electrode placed over the tail fin.

For recordings larvae were first paralyzed using alpha-bungarotoxin (1 mg/mL in extracellular solution containing 134 mM NaCl, 2.9 mM KCl, 1.2 mM MgCl₂, 10 mM HEPES, 10 mM glucose, and 2.1 mM CaCl₂, adjusted to pH 7.8 with NaOH), immersed in extracellular solution containing anesthetic (MS-222), and secured left side up on a sylgard lined glass bottom dish using custom

etched tungsten pins through the notochord. To record the activity of the peripheral ventral rootlets, the skin over the muscle tissue at the midbody was removed using fine forceps. For access to spinal V2a neurons, the muscle tissue over one spinal segment was dissected away using a tool fashioned from sharpened tungsten wire. After the dissection, the preparation was rinsed and immersed in anesthetic-free extracellular solution. For cell-attached and whole cell recordings, standard-wall (1 mm outer diameter) borosilicate capillaries (Sutter Instrument) were pulled to resistances between 5–16 M Ω using a micropipette puller (Flaming/Brown; Sutter Instrument). For peripheral motor nerve recordings, electrodes were made by pulling a patch electrode, breaking the tip, and then heat polishing and bending it using a microforge (MF-830; Narishige) to offset the approach angle for better contact between the electrode and muscle tissue. Peripheral motor nerve recordings were made no more than two segments caudal to the site of cellular recording.

For current clamp recordings, patch electrodes were filled with intracellular solution containing (125–130 mM K-gluconate, 2–4 mM MgCl₂, 0.2–10 mM EGTA, 10 mM HEPES, 4 mM Na₂ATP, adjusted to pH 7.3 with KOH). For voltage clamp recordings, the intracellular solution consisted of 122 mM CsMeSO₃, 0.1–1 mM QX314-Cl, 1 mM TEA-Cl, 3 mM MgCl₂, 10 mM HEPES, 1 mM EGTA, and 4 mM Na₂-ATP. CsMeSO₃ and QX-314 Cl were included in the solution to block voltage dependent channels. Alexa Fluor 568 hydrazide (50 μ mol/l) was also added to the intracellular solution for the post-hoc confirmation of cell identity.

V2a neurons were targeted on an epifluorescent microscope (Axio Examiner) under a 40 $\times$ /1.0 NA water-immersion objective (W Plan-Apochromat; Zeiss) using a CCD camera (Rolera-XR; QImaging). The electrode was positioned using a motorized micromanipulator (MP-225; Sutter Instrument or Patchstar, Scientifica), while positive pressure was maintained using a pneumatic transducer (Model DPM-1B; Bio-Tek Instruments) to prevent the tip from clogging as the neuron was approached. The neuron was held at -65 mV during G Ω seal formation. Once the seal was achieved, for voltage-clamp experiments cell attached recordings were performed to obtain the recruitment patterns of V2a neurons during fictive escape swimming before breaking into the neuron to record synaptic currents. A 5 mV hyperpolarizing pulse was then administered at a holding potential of -65 mV, and the steady state current injected during the pulse was used to compute input resistance according to Ohm's law.

Excitatory currents were clamped at the reversal potential for inhibition calculated from our solutions, which is -75 mV and inhibitory currents were clamped at the reversal potential for excitation, which is +10 mV. These values were corrected for a calculated junction potential of -11 mV. Thus, neurons were alternately held at the corrected potentials of -64 mV for EPSCs and 21 mV for IPSCs. To monitor series resistance throughout the experiment, recordings were alternately performed without and with 60% series resistance compensation. Standard corrections for bridge balance and electrode capacitance were applied for current clamp recordings. Whole-cell electrical signals were filtered at 10 kHz and digitized at 100 kHz at a gain of 10 (feedback resistor, 500 M Ω). Extracellular signals from the peripheral motor nerves were amplified at a gain of 1000 and digitized with low-frequency and high-frequency cutoffs set at 300 and 5000 Hz, respectively.

Imaging

After each electrophysiological recording, a series of epifluorescent and differential interference contrast images were captured using a CCD camera (Rolera-XR) and QCapture Suite imaging software (QImaging) to track the axonal trajectories of V2a neurons and confirm their morphology. Images were later analyzed in ImageJ.

Data analysis

Soma size and dorsoventral locations of individually labeled V2a neurons were measured in ImageJ. Dorsoventral (DV) and mediolateral (ML) positions of V2a neurons in Tg[Chx10:GFP] were computed by obtaining the relevant positional coordinates from confocal images in Imaris (Oxford

Instruments). Briefly, the spot detection tool was used to obtain the coordinates of the centers of all somata and orthogonal slicers were used in conjunction with measurement points to extract the coordinates of the dorsoventral and lateral boundaries of the spinal cord as well as the central canal. The positions of the somata were then respectively normalized against the positions of the dorsoventral and mediolateral boundaries to get the final DV and ML positions. V2a somata were classified into two groups based on the intensity of GFP expression. The GFP fluorescence intensity of V2a somata was exported from Imaris, and its density distribution was plotted in R. Because the density distribution of GFP fluorescence intensity was bimodal, the minimum between the two modes of the distribution was used as a cut off to split V2a neurons into classes of low and high GFP expression.

Whole cell and extracellular recordings were annotated in Dataview to label individual spikes, ventral rootlet bursts and swimming episodes. The onset and duration of spikes, ventral root bursts and swimming episodes were then batch exported into text files, which were then read into MATLAB (Mathworks) for further analysis. Instantaneous spike frequencies are the inverse of the period between two successive action potentials. The recruitment patterns of V2a neurons were quantified by computing the probability of observing spikes for all cycles present within each 5 Hz ventral root burst frequency bin from 10 Hz to 65 Hz. The recruitment probability was then plotted as a function of ventral root burst frequency. The V2a neurons with an upward sloping recruitment reliability plot were classified as fast and those with a downward sloping plot were categorized as slow (cf. trendlines in [Fig. 4E](#) )

Maximum excitatory and inhibitory currents reflect the maximum current received at any ventral root burst frequency. To compare frequency dependent changes in synaptic currents across neuron types, they were normalized against the maximum synaptic current the neuron received. Normalized currents were then averaged in four discrete frequency bins spanning carangiform and anguilliform swimming modes (15-25, 25-35, 35-45, 45+ Hz). To analyze the phasic timing of spiking and synaptic currents, they were normalized to the period of the corresponding cycle, such that the centers of successive motor bursts marked phase 0 and phase 1. Synaptic currents arriving between 0.75 to 0.25, were considered in-phase and those between 0.25 and 0.75 considered anti-phase. To analyze the relative timing of excitation and inhibition ([Fig. 6A, D](#)  and [E](#)), we normalized amplitudes to the maximum current received in each cycle, such that the final values of normalized current ranged between 0 and 1. Currents were then averaged for all cycles present within the four frequency bins

To quantify peaks in synaptic currents corresponding to the phasic inputs received during a swim cycle, current traces were low pass filtered at the cutoff frequency of 1000 Hz using the butterworth (second order) and zero phase digital filters in MATLAB. The latter was used to prevent the raw current values from undergoing a phase shift. Peaks were detected using the `findpeaks()` function in MATLAB. Counts of V2a spikes, EPSC's and IPSC's were plotted on a circular axis from phase 0 to 1. Data visualization and statistical analysis were performed in R.

Statistics

All data were first tested for normality to select the appropriate statistical test. None of the data were normally distributed, so nonparametric tests were used to assess statistical significance. The difference between the density distributions of mediolateral and dorsoventral somatic positions was tested using the Two-sample Kolmogorov-Smirnov test. Correlation between peak synaptic current and input resistance was tested using Spearman's rank correlation test. Comparisons between two groups of paired or unpaired data were performed using the Wilcoxon signed-rank test and the Wilcoxon rank sum test, respectively. For circular plots, Watson's two-sample test of homogeneity was used to determine whether the spike and post synaptic current distributions in two conditions were significantly different from each other. Statistically significant differences are

denoted as follows: $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. Note that it was not always possible to collect cell-attached data and voltage-clamp recordings of excitation and inhibition in the same cells, so n numbers reported in the text can vary between conditions.

Conflict of interest

The authors declare no conflict of interest.

Acknowledgements

Support provided by NIH R21 NS125187 and NIH R21 NS125207.

References

- Alvarez FJ, Jonas PC, Sapir T, Hartley R, Berrocal MC, Geiman EJ, Todd AJ, Goulding M (2005) **Postnatal phenotype and localization of spinal cord V1 derived interneurons** *J Comp Neurol* **493**:177–192
- Ausborn J, Shevtsova NA, Caggiano V, Danner SM, Rybak IA (2019) **Computational modeling of brainstem circuits controlling locomotor frequency and gait** *Elife* **8**
- Bello-Rojas S, Istrate AE, Kishore S, McLean DL (2019) **Central and peripheral innervation patterns of defined axial motor units in larval zebrafish** *J Comp Neurol* **527**:2557–2572
- Bhatt DH, McLean DL, Hale ME, Fetcho JR (2007) **Grading movement strength by changes in firing intensity versus recruitment of spinal interneurons** *Neuron* **53**:91–102
- Bhattacharyya K, McLean DL, MacIver MA (2021) **Intersection of motor volumes predicts the outcome of ambush predation of larval zebrafish** *J Exp Biol* **224**
- Brocard F, Tazerart S, Vinay L (2010) **Do pacemakers drive the central pattern generator for locomotion in mammals?** *Neuroscientist* **16**:139–155
- Brown TG (1911) **The intrinsic factors in the act of progression in the mammal** *Proc R Soc B – Biol Sci* **84**:308–319
- Buchanan JT (2018) **Swimming rhythm generation in the caudal hindbrain of the lamprey** *J Neurophysiol* **119**:1681–1692
- Buchanan JT, McPherson DR (1995) **The neuronal network for locomotion in the lamprey spinal cord: evidence for the involvement of commissural interneurons** *J Physiol Paris* **89**:221–233
- Budick SA, O'Malley DM (2000) **Locomotor repertoire of the larval zebrafish: swimming, turning and prey capture** *J Exp Biol* **203**:2565–2579
- Cangiano L, Grillner S (2018) **Commentary: elimination of left-right reciprocal coupling in the adult lamprey spinal cord abolishes the generation of locomotor activity** *Front Neural Circuits* **12**
- D'Elia KP *et al.* (2023) **Determinants of motor neuron functional subtypes important for locomotor speed** *Cell Rep* **42**
- Di Santo V, Goerig E, Wainwright DK, Akanyeti O, Liao JC, Castro-Santos T, Lauder GV (2021) **Convergence of undulatory swimming kinematics across a diversity of fishes** *Proc Natl Acad Sci U S A* **118**
- Dougherty KJ, Kiehn O (2010) **Firing and cellular properties of V2a interneurons in the rodent spinal cord** *J Neurosci* **30**:24–37
- El Manira A (2023) **Modular circuit organization for speed control of locomotor movements** *Curr Opin Neurobiol* **82**

Enoka RM, Duchateau J (2017) **Rate coding and the control of muscle force** *Cold Spring Harb Perspect Med* **7**

Feldman JL, Terjung R (1986) **Neurophysiology of breathing in mammals** *Handbook of Physiology*

Gosgnach S, Lanuza GM, Butt SJ, Saueressig H, Zhang Y, Velasquez T, Riethmacher D, Callaway EM, Kiehn O, Goulding M (2006) **V1 spinal neurons regulate the speed of vertebrate locomotor outputs** *Nature* **440**:215–219

Grillner S (1974) **On the generation of locomotion in the spinal dogfish** *Exp Brain Res* **20**:459–470

Grillner S, Terjung R (1981) **Control of locomotion in bipeds, tetrapods, and fish** *Handbook of Physiology*

Grillner S (1985) **Neurobiological bases of rhythmic motor acts in vertebrates** *Science* **228**:143–149

Grillner S, Kozlov A (2021) **The CPGs for limbed locomotion-facts and fiction** *Int J Mol Sci* **22**

Guertin PA (2009) **The mammalian central pattern generator for locomotion** *Brain Res Rev* **62**:45–56

Hayashi M, Hinckley CA, Driscoll SP, Moore NJ, Levine AJ, Hilde KL, Sharma K, Pfaff SL (2018) **Graded arrays of spinal and supraspinal V2a interneuron subtypes underlie forelimb and hindlimb motor control** *Neuron* **97**:869–884

Hoff KV, Wassersug RJ (2000) **Tadpole locomotion: axial movement and tail functions in a largely vertebraeless vertebrate** *American Zoologist* **40**:62–76

Hultborn H, Lindstrom S, Wigstrom H (1979) **On the function of recurrent inhibition in the spinal cord** *Exp Brain Res* **37**:399–403

Jay M, MacIver MA, McLean DL (2023) **Spinal basis of direction control during locomotion in larval zebrafish** *J Neurosci* **43**:4062–4074

Kimura Y, Higashijima SI (2019) **Regulation of locomotor speed and selection of active sets of neurons by V1 neurons** *Nat Commun* **10**

Kimura Y, Okamura Y, Higashijima S (2006) **alx, a zebrafish homolog of Chx10, marks ipsilateral descending excitatory interneurons that participate in the regulation of spinal locomotor circuits** *J Neurosci* **26**:5684–5697

Kimura Y, Satou C, Fujioka S, Shoji W, Umeda K, Ishizuka T, Yawo H, Higashijima S (2013) **Hindbrain V2a neurons in the excitation of spinal locomotor circuits during zebrafish swimming** *Curr Biol* **23**:843–849

Kishore S, Bagnall MW, McLean DL (2014) **Systematic shifts in the balance of excitation and inhibition coordinate the activity of axial motor pools at different speeds of locomotion** *J Neurosci* **34**:14046–14054

- Kozlov A, Huss M, Lansner A, Koteleski JH, Grillner S (2009) **Simple cellular and network control principles govern complex patterns of motor behavior** *Proc Natl Acad Sci U S A* **106**:20027–20032
- Kyriakatos A, Mahmood R, Ausborn J, Porres CP, Buschges A, El Manira A (2011) **Initiation of locomotion in adult zebrafish** *J Neurosci* **31**:8422–8431
- Li WC, Roberts A, Soffe SR (2010) **Specific brainstem neurons switch each other into pacemaker mode to drive movement by activating NMDA receptors** *J Neurosci* **30**:16609–16620
- Linden H, Berg RW (2021) **Why firing rate distributions are important for understanding spinal central pattern generators** *Front Hum Neurosci* **15**
- Marder E, Calabrese RL (1996) **Principles of rhythmic motor pattern generation** *Physiol Rev* **76**:687–717
- Marder E, Bucher D (2001) **Central pattern generators and the control of rhythmic movements** *Curr Biol* **11**:R986–996
- McClellan AD (2018) **Response: commentary: elimination of left-right reciprocal coupling in the adult lamprey spinal cord abolishes the generation of locomotor activity** *Front Neural Circuits* **12**
- McClellan AD, Pale T, Messina JA, Buso S, Shebib A (2016) **Similarities and differences for swimming in larval and adult lampreys** *Physiol Biochem Zool* **89**:294–312
- McCrea DA, Rybak IA (2008) **Organization of mammalian locomotor rhythm and pattern generation** *Brain Res Rev* **57**:134–146
- McHenry MJ, Lauder GV (2006) **Ontogeny of form and function: locomotor morphology and drag in zebrafish (*Danio rerio*)** *J Morphol* **267**:1099–1109
- McLean DL, Fetcho JR (2009) **Spinal interneurons differentiate sequentially from those driving the fastest swimming movements in larval zebrafish to those driving the slowest ones** *J Neurosci* **29**:13566–13577
- McLean DL, Dougherty KJ (2015) **Peeling back the layers of locomotor control in the spinal cord** *Curr Opin Neurobiol* **33**:63–70
- McLean DL, Fan J, Higashijima S, Hale ME, Fetcho JR (2007) **A topographic map of recruitment in spinal cord** *Nature* **446**:71–75
- McLean DL, Masino MA, Koh IY, Lindquist WB, Fetcho JR (2008) **Continuous shifts in the active set of spinal interneurons during changes in locomotor speed** *Nat Neurosci* **11**:1419–1429
- Menelaou E, McLean DL (2012) **A gradient in endogenous rhythmicity and oscillatory drive matches recruitment order in an axial motor pool** *J Neurosci* **32**:10925–10939
- Menelaou E, McLean DL (2019) **Hierarchical control of locomotion by distinct types of spinal V2a interneurons in zebrafish** *Nat Commun* **10**

- Menelaou E, VanDunk C, McLean DL (2014) **Differences in the morphology of spinal V2a neurons reflect their recruitment order during swimming in larval zebrafish** *J Comp Neurol* **522**:1232–1248
- Menelaou E, Kishore S, McLean DL (2022) **Mixed synapses reconcile violations of the size principle in zebrafish spinal cord** *Elife* **11**
- Messina JA, St Paul A, Hargis S, Thompson WE, McClellan AD (2017) **Elimination of left-right reciprocal coupling in the adult lamprey spinal cord abolishes the generation of locomotor activity** *Front Neural Circuits* **11**
- Moult PR, Cottrell GA, Li WC (2013) **Fast silencing reveals a lost role for reciprocal inhibition in locomotion** *Neuron* **77**:129–140
- Muller UK, van Leeuwen JL (2004) **Swimming of larval zebrafish: ontogeny of body waves and implications for locomotory development** *J Exp Biol* **207**:853–868
- Myers P, Eisen J, Westerfield M (1986) **Development and axonal outgrowth of identified motoneurons in the zebrafish** *J Neurosci* **6**:2278–2289
- Parker D (2017) **The lesioned spinal cord is a “new” spinal cord: evidence from functional changes after spinal injury in lamprey** *Front Neural Circuits* **11**
- Patterson BW, Abraham AO, MacIver MA, McLean DL (2013) **Visually guided gradation of prey capture movements in larval zebrafish** *J Exp Biol* **216**:3071–3083
- Picton LD, Bjornfors ER, Fontanel P, Pallucchi I, Bertuzzi M, El Manira A (2022) **Developmental switch in the function of inhibitory commissural V0d interneurons in zebrafish** *Curr Biol* **32**:3515–3528
- Ryczko D, Dubuc R, Cabelguen JM (2010) **Rhythmogenesis in axial locomotor networks: an interspecies comparison** *Prog Brain Res* **187**:189–211
- Satou C, Sugiyama T, Uemura Y, Shimazaki T, Zmarz P, Kimura Y, Higashijima SI (2020) **Functional diversity of glycinergic commissural inhibitory neurons in larval zebrafish** *Cell Rep* **30**:3036–3050
- Smith JL, Betts B, Edgerton VR, Zernicke RF (1980) **Rapid ankle extension during paw shakes: selective recruitment of fast ankle extensors** *J Neurophysiol* **43**:612–620
- Song J, Pallucchi I, Ausborn J, Ampatzis K, Bertuzzi M, Fontanel P, Picton LD, El Manira A (2020) **Multiple rhythm-generating circuits act in tandem with pacemaker properties to control the start and speed of locomotion** *Neuron* **105**:1048–1061
- Soto AP, McHenry MJ (2020) **Pursuit predation with intermittent locomotion in zebrafish** *J Exp Biol* **223**
- Talpalar AE, Bouvier J, Borgius L, Fortin G, Pierani A, Kiehn O (2013) **Dual-mode operation of neuronal networks involved in left-right alternation** *Nature* **500**:85–88
- Thorsen DH, Cassidy JJ, Hale ME (2004) **Swimming of larval zebrafish: fin-axis coordination and implications for function and neural control** *J Exp Biol* **207**:4175–4183

Wiggin TD, Anderson TM, Eian J, Peck JH, Masino MA (2012) **Episodic swimming in the larval zebrafish is generated by a spatially distributed spinal network with modular functional organization** *J Neurophysiol* **108**:925–934

Zhang J, Lanuza GM, Britz O, Wang Z, Siembab VC, Zhang Y, Velasquez T, Alvarez FJ, Frank E, Goulding M (2014) **V1 and V2b interneurons secure the alternating flexor-extensor motor activity mice require for limbed locomotion** *Neuron* **82**:138–150

Zhong G, Sharma K, Harris-Warrick RM (2011) **Frequency-dependent recruitment of V2a interneurons during fictive locomotion in the mouse spinal cord** *Nat Commun* **2**

Zhong G, Droho S, Crone SA, Dietz S, Kwan AC, Webb WW, Sharma K, Harris-Warrick RM (2010) **Electrophysiological characterization of V2a interneurons and their locomotor-related activity in the neonatal mouse spinal cord** *J Neurosci* **30**:170–182

Ziskind-Conhaim L, Hochman S (2017) **Diversity of molecularly defined spinal interneurons engaged in mammalian locomotor pattern generation** *J Neurophysiol* **118**:2956–2974

Article and author information

Moneeza A. Agha

Department of Neurobiology, Northwestern University, Evanston, IL 60201, Interdisciplinary Biological Sciences Graduate Program, Northwestern University, Evanston, IL 60201
ORCID iD: [0000-0001-7868-8325](https://orcid.org/0000-0001-7868-8325)

Sandeep Kishore

Department of Neurobiology, Northwestern University, Evanston, IL 60201

David L. McLean

Department of Neurobiology, Northwestern University, Evanston, IL 60201, Interdisciplinary Biological Sciences Graduate Program, Northwestern University, Evanston, IL 60201
For correspondence: david-mclean@northwestern.edu
ORCID iD: [0000-0001-6337-2301](https://orcid.org/0000-0001-6337-2301)

Copyright

© 2024, Agha et al.

This article is distributed under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/), which permits unrestricted use and redistribution provided that the original author and source are credited.

Editors

Reviewing Editor

Vatsala Thirumalai

National Centre for Biological Sciences, Bangalore, India

Senior Editor

Albert Cardona

University of Cambridge, Cambridge, United Kingdom

Reviewer #1 (Public Review):**Summary:**

In this very interesting study, Agha and colleagues show that two types of Chx10-positive neurons (V2a neurons) have different anatomical and electrophysiological properties and receive distinct patterns of excitatory and inhibitory inputs as a function of speed during fictive swimming in the larval zebrafish. Using single-cell fills they show that one cell type has a descending axon ("descending V2as"), while the other cell type has both a descending axon and an ascending axon ("bifurcating V2as"). In the Chx10:GFP line, descending V2as display strong GFP labeling, while bifurcating V2as display weak GFP labeling. The bifurcating V2as are located more laterally in the spinal cord. These two cell types have different electrophysiological properties as revealed by patch-clamp recordings. Positive current steps indicated that descending V2as comprise tonic spiking or bursting neurons. Bifurcating V2as comprise chattering or bursting neurons. The two types of V2a neurons display different recruitment patterns as a function of speed. Descending tonic and bifurcating chattering neurons are recruited at the beginning of the swimming bout, at fast speeds (swimming frequency above 30 Hz). Descending bursting neurons were preferentially recruited at the end of swimming bouts, at low speeds (swimming frequency below 30 Hz), while bifurcating bursting neurons were recruited for a broader swimming frequency range. The two types of V2a neurons receive distinct patterns of excitatory and inhibitory inputs during fictive locomotion. In descending V2as, when speed increases: i) excitatory conductances increase in fast neurons and decrease in slow neurons; ii) inhibitory conductances increase in fast neurons and increase in slow neurons. In bifurcating V2as, when speed increases: i) excitatory conductances increase in fast neurons but do not change in slow neurons; ii) inhibitory conductances increase in fast neurons and do not change in slow neurons. The timing of excitatory and inhibitory inputs was then studied. In descending V2as, fast neurons receive excitatory and inhibitory inputs that are in anti-phase with low contrast in amplitude and are both broadly distributed over the phase. The slow neurons receive two peaks of inhibition, one in anti-phase with the excitatory inputs and another just after the excitation. In bifurcating V2as, fast neurons receive two peaks of inhibition, while slow ones receive anti-phase inhibition.

Strengths:

This study focuses on the diversity of V2a neurons in zebrafish, an interesting cell population playing important roles in locomotor control and beyond, from fish to mammals. The authors provide compelling evidence that two subtypes of V2as show distinct anatomical, electrophysiological, and speed-dependent spiking activity, and receive distinct synaptic inputs as a function of speed. This opens the door to future investigation of the inputs and outputs of these neurons. Finding ways to activate or inhibit specifically these cells would be very helpful in the years to come.

Weaknesses:

No major weakness was detected. The experiments were carefully done, and the data were of high quality.

<https://doi.org/10.7554/eLife.94349.1.sa2>

Reviewer #2 (Public Review):**Summary:**

Animals exhibit different speeds of locomotion. In vertebrates, this is thought to be implemented by different groups of spinal interneurons and motor neurons. A fundamental assumption in the field has been that neural mechanisms that generate and sustain the rhythm at different locomotor speeds are the same. In this study, the authors challenge this

view. Using rigorous *in vivo* electrophysiology during fictive locomotion combined with genetics, the authors provide a detailed analysis of cellular and synaptic properties of different subtypes of spinal V2a neurons that play a crucial role in rhythm generation. Importantly, they are able to show that speed-related subsets of V2a neurons have distinct cellular and synaptic properties and may utilize different mechanisms to implement different locomotor speeds.

Strengths:

The authors fully utilize the zebrafish model system and solid electrophysiological analyses to study the active and passive properties of speed-related V2a subsets. Identification of the V2a subtype is based directly on their recruitment at different locomotor speeds and not on indirect markers like soma size, D-V position etc. Throughout the article, the authors have cleverly used standard electrophysiological tests and analysis to tease out different neuronal properties and link it to natural activity. For example, in Figures 2 and 4, the authors make comparisons of V2a spiking with current steps and during fictive swims showing spike rates measured with current steps are physiologically relevant and observed during natural recruitment. The experiments done are rigorous and well-controlled.

Weaknesses:

The authors claim that a primary result of their study is that reciprocal inhibition is important for rhythmogenesis at fast speeds while recurrent inhibition is key at slow speeds. This is shown in Figure 6, however, the authors do not show any statistical tests for this claim. The authors also do not show any conclusive evidence that reciprocal inhibition is required for rhythmogenesis at fast speeds and vice versa for slow speeds. Additional experiments or modeling studies that conclusively show the necessity of these different inhibitory sources to the generation of different rhythms would be needed to strengthen this claim.

The authors do a great job of teasing out cellular and synaptic properties in the different V2a subsets, however, it is not clear if or how these match the final output. For example, V2aD neurons are tonic or bursting for fast and slow speeds respectively but it is not intuitive how these cellular properties would influence phasic excitation and inhibition these neurons receive.

It is not clear from the discussion why having different mechanisms of rhythm generation at different speeds could be an important circuit design. The authors use anguilliform and carangiform modes of swimming to denote fast and slow speeds but there are differences in these movements other than speed, like rostrocaudal coordination. The frequency and pattern of these movements are linked and warrant more discussion.

<https://doi.org/10.7554/eLife.94349.1.sa1>

Reviewer #3 (Public Review):

The manuscript by Agha et al. explores mechanisms of rhythmicity in V2a neurons in larval zebrafish. Two subpopulations of V2a neurons are distinguishable by anatomy, connectivity, level of GFP, and speed-dependent recruitment properties consistent with V2a neurons involved in rhythm generation and pattern formation. The descending neurons proposed to be consistent with rhythm-generating neurons are active during either slow or fast locomotion, and their firing frequencies during current steps are well matched with the swim frequency they firing during. The bifurcating (patterning neurons) are active during a broader swim frequency range unrelated to their firing during current steps. All of the V2a neurons receive strong inhibitory input but the phasing of this input is based on neuronal type and swim speed when the neuron is active, with prominent in-phase inhibition in slow descending V2a neurons and bifurcating V2a neurons active during fast swimming. Antiphase inhibition is observed in all V2a neurons but it is the main source of rhythmic

inhibition in fast descending V2a neurons and bifurcating neurons active during slow swimming. The authors suggest that properties supporting rhythmic bursting are not directly related to locomotor speed but rather to functional neuronal subtypes.

This is a well-written paper with many strengths including the rigorous approach. Many parameters, including projection pattern, intracellular properties, inhibition received, and activity during slow/fast swimming were obtained from the same neuron. This links up very well with prior data from the lab on cell position, birth order, morphology/projections, and control of MN recruitment to provide a comprehensive overview of the functioning of V2a interneuronal populations in the larval zebrafish. The overall conclusions are well supported by the data. Weaknesses are relatively minor and were largely related to terminology for some of the secondary conclusions.

1. The assumption is made that all in-phase inhibition is recurrent and out-of-phase inhibition is reciprocal. The latter is likely true but the definition of recurrent may be a bit loose as could be multisegmental feed-forward inhibition as well.

2. In a few places, it is mentioned that the properties of the V2a-D neurons are consistent with pacemakers. This could be true of both the V2a-D and -B neurons that burst in response to depolarizing steps but the properties of the remaining (fast) V2a-D neurons do not seem to be consistent with pacemakers, based on the properties shown. Tonic firing at a frequency related to the locomotor speed the neuron is active during and strong antiphase inhibition may instead suggest a stronger network component driving the rhythmicity.

<https://doi.org/10.7554/eLife.94349.1.sa0>